Hongsong Wang（王洪松）

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BACKGROUND

National University of Singapore July 2018 – June 2019
Post-doctoral Research Fellow

Institute of Automation, Chinese Academy of Sciences Sept 2013 – June 2018
Ph.D. in National Laboratory of Pattern Recognition, Supervisor: Prof. Liang Wang
Thesis title: Modelling Motion Evolution for Human Action Recognition

Huazhong University of Science and Technology Sept 2009 – June 2013
B.Sc. in Department of Control Science and Engineering
Thesis title: Content Based Image Retrieval Based on Deep Learning

SKILLS AND INTERESTS

Computer Vision: Action Recognition, Action Prediction, Action Detection, Action Proposal, Video Classification, Human Motion Prediction.

Machine Learning: Deep Learning, Sparse Coding, Transfer Learning, Multi-task Learning, Bag-of-words Model.

Proficient in Python, Matlab.
Familiar with Pytorch, Tensorflow, Caffe, Theano, OpenCV.

PUBLICATIONS

Submitted Papers

[R1] **Hongsong Wang**, Jiashi Feng, PVRED: A Position-Aware Velocity-Assisted Recurrent Encoder-Decoder for Human Motion Prediction.

[R2] **Hongsong Wang**, Jiashi Feng, Delving into 3D Action Anticipation from Streaming Videos.

Journal Papers

[J1] **Hongsong Wang,** Liang Wang, Beyond Joints: Learning Representations from Primitive Geometries for Skeleton-based Action Recognition and Detection, IEEE Transactions on Image Processing (**TIP**), 2018.

[J2] **Hongsong Wang**, Liang Wang, Learning Content and Style: Joint Action Recognition and Person Identification from Human Skeletons, Pattern Recognition (**PR**), 2018.

[J3] **Hongsong Wang**, Liang Wang, Cross-Agent Action Recognition, IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2018.

Conference Papers

[C1] **Hongsong Wang,** Liang Wang, Modeling Temporal Dynamics and Spatial Configurations of Actions Using Two-Stream Recurrent Neural Networks, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2017.

[C2] **Hongsong Wang,** Liang Wang, Learning Robust Representations using Recurrent Neural Networks for Skeleton based Action Classification and Detection, IEEE International Conference on Multimedia and Expo Workshop on Large Scale 3D Human Activity Analysis Challenge in Depth Videos (**ICMEW Oral**), 2017.

[C3] **Hongsong Wang**, Wei Wang, Liang Wang, How Scene Implies Action in Realistic Videos?, IEEE International Conference on Image Processing (**ICIP oral**), 2016.

[C4] **Hongsong Wang**, Wei Wang, Liang Wang, Hierarchical Motion Evolution for Action Recognition, IEEE Asian Conference on Pattern Recognition (**ACPR oral**), 2015.

CONTESTS

Video Tagging Grand Challenge in Research and Application in Computer Vision (**RACV 2016**): **1st** place. (DIGCASIA: **Hongsong Wang**, Yong Du, Liang Wang)

Large Scale 3D Human Activity Analysis Challenge in Depth Videos in 13th Asian Conference on Computer Vision (**ACCV 2016**) Workshop: **2nd** place. (YANQIAN: **Hongsong Wang**, Liang Wang)

Recognition Track for Large Scale 3D Human Activity Analysis Challenge in Depth Videos in 18th International Conference on Multimedia & Expo (**ICME 2017**) Workshop: **2nd** place. (DIGCASIA: **Hongsong Wang**, Yuqi Zhang, Liang Wang)

Detection Track for Large Scale 3D Human Activity Analysis Challenge in Depth Videos in 18th International Conference on Multimedia & Expo (**ICME 2017**) Workshop: **2nd** place. (DIGCASIA: **Hongsong Wang**, Yuqi Zhang, Liang Wang)

HONORS AND AWARDS

- National First Prize in China Undergraduate Mathematical Contest in Modeling (CUMCM), 2011.
- Meritorious Winner in Mathematical Contest in Modeling (MCM) of American Undergraduate Students, 2012.
- National Endeavor Scholarship, Ministry of Education, P.R. China, 2010.
- Rockwell Automation Scholarship, 2011.
- Title of Self-Reliant Model in Department of Automation, 2012.
- Excellent Postgraduate of Huazhong University of Science and Technology, 2013.
- Excellent Student of University of Chinese Academy of Science, 2017.
- Dai Ruwei First Prize Scholarship, Institute of Automation of the Chinese Academy of Sciences (CASIA), 2018.
- CSIG Doctoral Dissertation Nomination Award, China Society of Image and Graphics (CSIG), 2018.

ACADEMIC ACTIVITIES

- The 3rd IAPR Asian Conference on Pattern Recognition (**ACPR**), November 3-6, 2015, Kuala Lumpur, Malaysia. (Oral presentation)
- The 2016 Symposium on Research and Application in Computer Vision, September 18-20, 2016, Shanghai, China. (A talk delivered)
- IEEE International Conference on Multimedia and Expo (**ICME**), July 10-14, 2017, Hong Kong. (Oral presentation)
- Reviewer of IEEE Transactions on Cybernetics.
- Reviewer of IEEE Access.
- Reviewer of Asian Conference on Computer Vision (**ACCV**), 2018.

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教育背景

- **新加坡国立大学，电子与计算机工程系** 2018.07–2019.06
  博士后

- **中科院自动化所，模式识别国家重点实验室** 2013.09–2018.07
  工学博士，专业：模式识别与智能系统，导师：王亮
  博士论文：基于动作演变的人体行为识别研究

- **华中科技大学，控制科学与工程学院** 2009.09–2013.07
  工学学士，专业：自动化
  学士论文: 基于深度学习的图像检索

专业技能

- **计算机视觉：**行为识别，行为检测，行为预测，视频分类，人体动作预测。
- **机器学习：**深度学习，稀疏编码，迁移学习，多任务学习，视觉词典模型。
- 熟练掌握 Python，Matlab，C/C++，Linux 操作系统。
- 熟练掌握工具 Pytorch，Tensorflow，Caffe，Theano，OpenCV。

会议及期刊论文

- **Hongsong Wang**, Jiashi Feng, PVRED: A Position-Aware Velocity-Assisted Recurrent Encoder-Decoder for Human Motion Prediction. (投稿)
- **Hongsong Wang**, Jiashi Feng, Delving into 3D Action Anticipation from Streaming Videos. (投稿)
- **Hongsong Wang**, Liang Wang, Beyond Joints: Learning Representations from Primitive Geometries for Skeleton-based Action Recognition and Detection, IEEE Transactions on Image Processing (**TIP**), 2018. (**IF: 4.8**)
- **Hongsong Wang**, Liang Wang, Learning Content and Style from Human Skeletons with Multi-Task Recurrent Neural Networks, Pattern Recognition (**PR**), 2018. (**IF: 4.6**)
- **Hongsong Wang**, Liang Wang, Cross-Agent Action Recognition, IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2018. **(IF: 3.6**)
- **Hongsong Wang,** Liang Wang, Modeling Temporal Dynamics and Spatial Configurations of Actions Using Two-Stream Recurrent Neural Networks, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2017.
- **Hongsong Wang,** Liang Wang, Learning Robust Representations using Recurrent Neural Networks for Skeleton based Action Classification and Detection, IEEE International Conference on Multimedia and Expo Workshop on Large Scale 3D

Human Activity Analysis Challenge in Depth Videos (**ICMEW Oral**), 2017.

- **Hongsong Wang**, Wei Wang, Liang Wang, How Scene Implies Action in Realistic Videos?, IEEE International Conference on Image Processing (**ICIP oral**), 2016.
- **Hongsong Wang**, Wei Wang, Liang Wang, Hierarchical Motion Evolution for Action Recognition, IEEE Asian Conference on Pattern Recognition (**ACPR oral**), 2015.

国内外竞赛

- 2016 计算机视觉研究与应用创新论坛：爱奇艺视频标注技术挑战赛（专题竞赛），成绩：**第一名**，队名：DIGCASIA（**王洪松**，杜勇，王亮）。
- 2016 年亚洲计算机视觉会议专题讨论会（ACCV Workshop）：深度视频中的大规模人体行为分析挑战赛，成绩：**第二名**，队名：YANQIAN（**王洪松**，王亮）。
- 2017 年 IEEE 国际多媒体世博会专题讨论会（ICME Workshop）：深度视频中的大规模人体行为分析挑战赛。识别任务：**第二名**，检测任务：**第二名**，队名：DIGCASIA（**王洪松**，张宇琪，王亮）。

科研活动

- 2015 年 11 月在吉隆坡（马来西亚）参加亚洲模式识别学术会议，并做口头报告。
- 2016 年 8 月在北京参加中国人工智能大会（CCAI）。
- 2016 年 9 月在上海参加计算机视觉研究与应用创新论坛，并做口头报告。
- 2017 年 4 月在厦门参加视觉与学习青年学者研讨会（VALSE）。
- 2017 年 7 月在香港参加国际多媒体世博会议（ICME），并做口头报告。
- 审稿人：IEEE Transactions on Cybernetics，IEEE Access，Asian Conference on Computer Vision (ACCV)。

奖励与荣誉

- 2011 年高教社杯全国大学生数学建模竞赛全国一等奖。
- 2012 年美国大学生数学建模大赛一等奖。
- 2010 年国家励志奖学金，华中科技大学“校三好学生”荣誉称号。
- 2011 年罗克韦尔自动化奖学金。
- 2012 年控制科学与工程学院“自强标兵”称号，汇川奖学金。
- 2013 年华中科技大学优秀毕业生。
- 2017 年中国科学院大学“三好学生”荣誉称号。
- 2018 年中科院自动化研究所戴汝为一等奖学金。
- 2018 年中国图像图形学学会（CSIG）优秀博士学位论文奖提名奖。